Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

DEPARTMENT OF MECHATRONICS

VII SEMESTER B.TECH. (MECHATRONICS)

END SEMESTER EXAMINATIONS, Dec, 2021

SUBJECT: ROBOT PATH PLANNING AND MOBILE ROBOTICS [MTE 4061]

Time: 09:00 AM to 01:00 PM **Date:24/12/2021**

Time: 3 Hours **MAX. MARKS: 50**

Instructions to Candidates:

- Answer **ALL** the questions.

PART A		**30 Marks**				
		Marks	**CO**	**PO**	**LO**	**BL**
1	ROS was originally started at a. Stanford Artificial Intelligence Laboratory (SAIL) b. Carnegie Mellon Robotics lab c. General Robotics, Automation, Sensing and Perception (GRASP) Laboratory d. Distributed Robotics Laboratory, MIT	**1**	**1**	**1,2**	**1,2**	**1**
2	Define the command used for starting ROS Master through terminal command in ubuntu? a. $ roscore b. $ rosmaster c. $ rosver d. $ roservice	**1**	**2**	**1,2**	**1,2**	**1**
3	Identify the typical elements in typical ROS Package structure? e. src a. xml b. catkin c. build	**1**	**2**	**1,2**	**1,2**	**1**
4	Mobile robots have the capability to move around in their environment and are not fixed to one physical location? f. True a. False	**1**	**1**	**1,2**	**1,2**	**1**
5	Identify the type of underwater robots? g. AUV a. UAV b. AGV c. UGV	**1**	**1**	**1,2**	**1,2**	**1**

6	Predict the usages of cameras used in navigation? a. Analytical determination b. Texture Information c. Both a and b d. None of the above	**1**	**1**	**1,2**	**1,2**	**3**
7	The computational problem of constructing or updating a map of an unknown environment is done by the method a. SLAM Technique b. Perception Technique c. Path Planning Technique d. Actuation	**1**	**2**	**1,2**	**1,2**	**1**
8	How an agent ought to take actions in an environment to maximize the notion of a long-term reward. a. Artificial Neural Networks b. Deep learning c. Reinforcement learning d. Genetic algorithms	**1**	**1**	**1,2**	**1,2**	**1**
9	The set of configurations that avoids collision with obstacles is called as. a. Free Space b. Target space c. Configuration space d. Manifold space	**1**	**2**	**1,2**	**1,2**	**1**
10	The robot's configuration as a point (usually electron) in a potential field that combines attraction to the goal, and repulsion from obstacles usually represented in? a. Sampling-based algorithms b. Artificial potential fields c. Grid-based search d. Reward-based algorithms	**1**	**2**	**1,2**	**1,2**	**1**
11	List out the computer vision challenges in real time? a. Viewpoint changes b. Material Property c. Both a and b d. None of the above	**1**	**1**	**1,2**	**1,2**	**1**
12	Intelligent suggest mission commands and planning will be done on which phase of Autonomous Execution? a. Cognition Level b. Perception Level c. Actuation Level d. Mapping and Localization Level	**1**	**1**	**1,2**	**1,2**	**1**
13	Mechanical complexity will be more in legged robots than in wheeled robots a. True b. False	**1**	**1**	**1,2**	**1,2**	**1**
14	Define the parameters required for a kinematic motion for analyzing its movement? a. Speed b. Mass c. Force d. Torque	**1**	**1**	**1,2**	**1,2**	**1**

15	Define which is a simple mathematical representation of physical system _____? a. Model or Mathematical Model b. Dynamical Model c. Kinematic System d. Geometric System	**1**	**CO1**	**1,2**	**1,2**	**1**
16	For executing the turtlesim in ROS, select the correct command from below: a. rosrun turtlesim turtlesim_node b. rosrun turtle turtle_node c. rosrun turtle_node turtlesim d. None of the above	**1**	**CO2**	**1,2**	**1,2**	**1**
17	Execute turtlesim node, identify the output of command - **rostopic type /turtle1/cmd_vel ?** a. geometry_msgs/Twist b. msg_mags/Twist c. cmd_vel/Twist d. None of the above	**1**	**CO2**	**1,2**	**1,2**	**1**
18	To show the frequency rate in Hz for publication which command is used in topics? a. rostopic hz b. rostopic fer c. rostopic fz d. none of the above	**1**	**CO2**	**1,2**	**1,2**	**1**
19	How many parameters does cmd_vel topic contain in ROS? a. 3 b. 4 c. 6 d. 2	**1**	**CO2**	**1,2**	**1,2**	**1**
20	Illustrate cross edges in Depth Fast Search algorithms. Select the relevant statement from the below. a. True b. False	**1**	**CO3**	**1,2**	**1,2**	**1**
21	In breadth fast search we will not do any exploration of adjacent vertex. Select the relevant statement from the below. a. True b. False	**1**	**CO3**	**1,2**	**1,2**	**1**
22	In graph-based method the collision free path is achieved by which of the following strategies a. Vertex Connection b. Equidistant from the boundaries c. Sampling trees d. Attractive fields	**1**	**CO3**	**1,2**	**1,2**	**1**
23	The configuration space and cartesian space of two 2-R manipulator is same in real time for implementing path planning. a. True b. False	**1**	**CO3**	**1,2**	**1,2**	**1**
24	The distance values produced by the grassfire algorithm indicate the smallest number of steps needed to move from each node to the goal. a. True	**1**	**CO4**	**1,2**	**1,2**	**1**

	b. False					
25	Graphs and tree structures in algorithms are similar. Select the relevant statement from the below. a. True b. False	**1**	**CO4**	**1,2**	**1,2**	**1**
26	The significance of the generalized kinematics model of the wheel mobile robot is to get the relation between Vector of angular velocities to the vector of input commands a. True b. False	**1**	**CO1**	**1,2**	**1,2**	**1**
27	Analyze, for a differential drive mobile robot what will be values of angles between roller axis to the Xci axis a. 0 b. 90 c. 210 d. 180	**1**	**CO1**	**1,2**	**1,2**	**3**
28	The point where the total of horizontal inertia and gravity forces equals zero while performing dynamic motion is called as a. ZMP (Zero Moment Point) b. DBM (Dynamic Balance Margin) c. SSP (Single Support Phase) d. DSP (Double Support Phase)	**1**	**CO1**	**1,2**	**1,2**	**1**
29	In graph-based method the collision free path is achieved by which of the following strategies a. Vertex Connection b. Equidistant from the boundaries c. Sampling trees d. Attractive fields	**1**	**CO3**	**1,2**	**1,2**	**1**
30	In Bug Algorithm, m-line is the line connecting from which point to which point? a. Ql to Qh b. Start to Ql c. Qh to Goal d. Start to Goal	**1**	**CO2**	**1,2**	**1,2**	**1**